Lesson Planning

Complete Lesson Plan Book

Class: Class 10

Subject: Chemistry

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Chemical Equilibrium

Learning Objectives

• Define reversible and irreversible reactions.

• Explain dynamic equilibrium with examples.

• State the law of mass action.

• Apply Le Chatelier's principle.

Key Concepts

• Reversible vs irreversible reactions

• Dynamic equilibrium: rates equal, concentrations constant

• Law of mass action; equilibrium constant Kc

• Le Chatelier's principle: effect of concentration, temperature, pressure

• Industrial example: Haber process

Teaching Methodology (Step by Step)

1. Show a see-saw in balance: forward and backward rates equal.
2. Demonstrate the NO2/N2O4 colour equilibrium if possible.
3. Explain Kc with a simple worked example.
4. Use Le Chatelier to predict shifts in given reactions.

Classroom Activities

• Predict equilibrium shifts for five given changes.

• Write the Kc expression for three reactions.

Assessment Questions

1. What is meant by dynamic equilibrium?
2. State Le Chatelier's principle.
3. What happens to an exothermic equilibrium when heated?
4. Write the Kc expression for N2 + 3H2 <=> 2NH3.

Homework

• Solve problems on equilibrium shifts.

• Write a note on the Haber process.

Time Allocation

5 periods of 40 minutes each

Unit 2: Acids, Bases and Salts

Learning Objectives

• Define acids and bases (Arrhenius concept).

• Explain the pH scale and its uses.

• Describe neutralization and salt formation.

• Classify salts as acidic, basic or neutral.

Key Concepts

• Acid: produces H+; Base: produces OH-

• Properties of acids and bases

• pH scale 0-14; indicators (litmus, phenolphthalein)

• Neutralization: acid + base -> salt + water

• Types of salts and their uses

Teaching Methodology (Step by Step)

1. Test lemon juice and soap with litmus: the colour hook.
2. Build the pH scale on the board with daily examples.
3. Demonstrate neutralization with an indicator colour change.
4. Classify given salts by the nature of their parent acid/base.

Classroom Activities

• Test five household substances with litmus paper.

• Arrange ten substances on a pH-scale chart.

Assessment Questions

1. Define an acid according to Arrhenius.
2. What is the pH of a neutral solution?
3. Write the equation of HCl + NaOH.
4. Name two indicators and their colour changes.

Homework

• Complete the exercise on pH and neutralization.

• Write uses of any five salts.

Time Allocation

6 periods of 40 minutes each

DM

Unit 3: Organic Chemistry

Learning Objectives

• Define organic chemistry and organic compounds.

• Explain catenation and isomerism.

• Classify organic compounds.

• Draw structures of simple organic molecules.

Key Concepts

• Organic: compounds of carbon (with H, O, N...)

• Catenation: carbon's chain-forming ability

• Isomerism: same formula, different structure

• Functional groups: -OH, -COOH, -CHO

• Homologous series concept

Teaching Methodology (Step by Step)

1. Ask why there are millions of carbon compounds: catenation.
2. Build methane, ethane models with balls and sticks.
3. Draw isomers of butane (C4H10) together.
4. Introduce functional groups with familiar examples (alcohol, acid).

Classroom Activities

• Draw structural formulas of the first five alkanes.

• Draw both isomers of butane.

Assessment Questions

1. Define organic chemistry.
2. What is catenation?
3. Draw the structural formula of ethane.
4. Name two functional groups with their formulas.

Homework

• Draw structures of methane to pentane.

• Write a note on isomerism with diagrams.

Time Allocation

6 periods of 40 minutes each

DM

Unit 4: Hydrocarbons

Learning Objectives

• Classify hydrocarbons as alkanes, alkenes, alkynes.

• Write general formulas and IUPAC names.

• Describe preparation and properties of methane and ethene.

• Distinguish alkanes from alkenes by chemical test.

Key Concepts

• Alkanes CnH2n+2: single bonds, saturated

• Alkenes CnH2n: double bond, unsaturated

• Alkynes CnH2n-2: triple bond

• IUPAC naming basics

• Bromine-water test for unsaturation

Teaching Methodology (Step by Step)

1. Compare models of ethane, ethene, ethyne side by side.

2. Teach IUPAC naming with the longest-chain rule.

3. Demonstrate the bromine-water test (teacher demo).

4. Discuss uses: natural gas, plastics from ethene.

Classroom Activities

• Name ten given structures by IUPAC rules.

• Draw alkanes, alkenes and alkynes up to C4.

Assessment Questions

1. Write the general formula of alkenes.

2. Name CH3-CH2-CH3 by IUPAC.

3. How do you test for an alkene?

4. Why are alkenes more reactive than alkanes?

Homework

• Complete the IUPAC naming exercise.

• Write properties and uses of methane.

Time Allocation

6 periods of 40 minutes each

DM

Unit 5: Biochemistry

Learning Objectives

• Define biochemistry and its scope.

• Describe carbohydrates: types and functions.

• Describe proteins: amino acids and functions.

• Explain the roles of lipids, vitamins and nucleic acids.

Key Concepts

• Carbohydrates: monosaccharides, disaccharides, polysaccharides

• Proteins: amino acids, peptide bond; enzymes

• Lipids: fats and oils; energy storage

• Vitamins and deficiency diseases

• Nucleic acids: DNA and RNA basics

Teaching Methodology (Step by Step)

1. Link to biology: nutrients are chemistry in action.
2. Test foods for starch (iodine) and protein (biuret) as demos.
3. Build the carbohydrate family tree on the board.
4. Match vitamins with deficiency diseases in a game.

Classroom Activities

• Test three foods for starch; record results.

• Make a chart: nutrient, source, function, deficiency.

Assessment Questions

1. Name the three types of carbohydrates.
2. What are amino acids?
3. Which vitamin deficiency causes scurvy?
4. Write one function each of proteins and lipids.

Homework

• Write the classification of carbohydrates with examples.

• Draw a labelled diagram of a DNA double helix.

Time Allocation

5 periods of 40 minutes each

Unit 6: The Atmosphere

Learning Objectives

• Describe the composition of air.

• Explain the layers of the atmosphere.

• Describe air pollutants and their sources.

• Explain the greenhouse effect and acid rain.

Key Concepts

• Air: nitrogen 78%, oxygen 21%, others

• Layers: troposphere, stratosphere, mesosphere, thermosphere

• Ozone layer and its depletion

• Pollutants: CO, SO2, NOx, particulates

• Greenhouse effect, global warming, acid rain

Teaching Methodology (Step by Step)

1. Ask why we don't feel the air's weight: composition intro.
2. Draw the atmospheric layers with their features.
3. Discuss the ozone hole story simply.
4. Debate: what can our city do about smog?

Classroom Activities

• Make a pie chart of air composition.

• List five ways to reduce air pollution.

Assessment Questions

1. Write the percentage of nitrogen and oxygen in air.
2. In which layer is the ozone found?
3. Name three air pollutants.
4. What is acid rain?

Homework

• Draw the atmospheric layers with labels.

• Write a paragraph on the greenhouse effect.

Time Allocation

5 periods of 40 minutes each

DM

Unit 7: Water

Learning Objectives

• Describe the properties of water.

• Explain hard and soft water.

• Describe methods to remove hardness.

• Discuss water pollution and its prevention.

Key Concepts

• Water: universal solvent; polarity idea

• Hard water: Ca^{2+} and Mg^{2+} salts

• Temporary vs permanent hardness

• Removal: boiling, washing soda, ion exchange

• Water pollutants and treatment

Teaching Methodology (Step by Step)

1. Lather test: soap in tap vs distilled water.
2. Explain hardness with the scale-in-kettle example.
3. Demonstrate removal of temporary hardness by boiling.
4. Discuss local water issues and solutions.

Classroom Activities

• Test water samples from different sources for hardness.

• Design a poster on saving water.

Assessment Questions

1. What causes hardness in water?
2. Distinguish between temporary and permanent hardness.
3. Name two methods to remove hardness.
4. Why is water called a universal solvent?

Homework

• Write methods of removing hardness with equations.

• Write a paragraph on water pollution.

Time Allocation

4 periods of 40 minutes each

DM

Unit 8: Chemical Industries

Learning Objectives

• Describe the manufacture of urea (fertilizer).

• Explain the Solvay process for sodium carbonate.

• Describe the uses of common industrial chemicals.

• Discuss the importance of chemical industries for Pakistan.

Key Concepts

• Fertilizers: urea manufacture outline

• Solvay process: Na_2CO_3 from brine and limestone

• Sulphuric acid: contact process outline

• Soaps and detergents basics

• Industrial safety and environment

Teaching Methodology (Step by Step)

1. Connect fertilizers to the wheat fields students see.
2. Draw flow-sheet diagrams of the processes step by step.
3. Compare soap and detergent with hard-water examples.
4. Discuss jobs and economy: why industries matter.

Classroom Activities

• Draw the flow sheet of urea manufacture.

• List ten products of chemical industries used at home.

Assessment Questions

1. Name the raw materials of the Solvay process.
2. Write one use of urea.
3. What is the difference between soap and detergent?
4. Name two chemical industries of Pakistan.

Homework

• Draw flow sheets of the Solvay process.

• Write a paragraph on the importance of chemical industries.

Time Allocation

5 periods of 40 minutes each

DM